

Lesson 23: The Wave Equation - Vibrations in 1D

Professor Dave Explains Differential Equations Master Series · Lesson 23

Video: The Wave Equation Part 1: Vibrations in 1D **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that the Wave Equation $u_{tt} = c^2 u_{xx}$ is the defining law of sound, string instruments, seismic waves, and light. When you pluck a guitar string, tension pulls each curved segment back toward equilibrium: curvature u_{xx} creates an accelerating force u_{tt} with propagation speed $c = \sqrt{T/\rho}$. Jean le Rond d'Alembert discovered that every disturbance splits into two counter-propagating traveling waves $f(x - ct) + g(x + ct)$, while separation of variables reveals musical standing harmonics.

2. Core Theoretical Framework & Rigorous Formulations

1. The 1D Wave Equation Formulation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad c = \sqrt{\frac{T}{\rho}}$$

Boundary conditions: $u(0, t) = 0, u(L, t) = 0$. Initial conditions: $u(x, 0) = f(x), u_t(x, 0) = g(x)$.

2. Standing Normal Modes of Vibration

$$u(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right) \right] \sin\left(\frac{n\pi x}{L}\right)$$

Natural angular frequencies: $\omega_n = \frac{n\pi c}{L}$. Fundamental frequency: $f_1 = \frac{c}{2L}$.

3. d'Alembert's Traveling Wave Formula

$$u(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds$$

3. Fully Solved Whiteboard Problems

Vibrations Problem: Plucked Guitar String Released from Rest

Problem Statement: A string of length $L = 1$ with propagation speed $c = 2$ is plucked into an initial triangle shape $f(x) = \begin{cases} 2x, & 0 \leq x \leq 1/2 \\ 2(1-x), & 1/2 \leq x \leq 1 \end{cases}$ and released from rest ($g(x) = 0$). Determine the normal mode coefficients.

Step 1: Evaluate Velocity Condition

Since released from rest: $g(x) = 0 \implies B_n = 0$ for all n .

Step 2: Formulate Modal Expansion

$$u(x, t) = \sum_{n=1}^{\infty} A_n \cos(2n\pi t) \sin(n\pi x).$$

Step 3: Integrate Fourier Coefficient A_n

$$A_n = 2 \int_0^1 f(x) \sin(n\pi x) dx = 2 \left[\int_0^{1/2} 2x \sin(n\pi x) dx + \int_{1/2}^1 2(1-x) \sin(n\pi x) dx \right].$$

Step 4: Execute Integration by Parts

Evaluating the integrals yields: $A_n = \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$.

Step 5: Parity Analysis of Normal Modes

For even $n = 2, 4, 6$: $\sin(n\pi/2) = 0 \implies A_n = 0$ (no even harmonics excited!). For odd n : $A_1 = \frac{8}{\pi^2}, A_3 = -\frac{8}{9\pi^2}, A_5 = \frac{8}{25\pi^2}$.

Step 6: Assemble Full Modal Motion

$$u(x, t) = \frac{8}{\pi^2} (\cos(2\pi t) \sin(\pi x) - \frac{1}{9} \cos(6\pi t) \sin(3\pi x) + \frac{1}{25} \cos(10\pi t) \sin(5\pi x) - \dots).$$

⚠ Common Exam Pitfall & Trapdoor Avoidance

When evaluating A_n , remember that plucking at the exact midpoint $x = 1/2$ creates perfect symmetry, causing all even harmonics (octaves) to cancel out completely.

4. Key Takeaways & Exam Strategy Formula Card

Core Takeaway & Exam Rule

The wave equation conserves mechanical energy indefinitely ($u_{tt} = c^2 u_{xx}$). Standing waves are linear superpositions of counter-propagating traveling waves.

- Video reference: [The Wave Equation Part 1: Vibrations in 1D](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.